Inorganic Cumulative Exam
March 26, 2004

You are an assistant professor in the first year of your career and eager for success. You are interested in the transition metal chemistry of carbonates. In your 83th reaction you mix two equivalents of 1,10-phenantholine (a chelate) with one equivalent of $FeCl_3$ followed by a large excess of sodium carbonate. You recover a bright red precipitate. Being eager for success you retire to your office and begin to prepare a manuscript for publication of your results. You immediately realize that you face two challenges when writing the paper. You need to provide an introductory paragraph stating the significance of your studies. You begin to ponder the importance of atmospheric CO_2 and its biological activity among other things. After getting something down on paper you begin to question the validity of your results and begin to ponder the question of how best to characterize the product (we will assume that Murphy was out to lunch and the compound is pure but not crystalline).

1). Provide an introductory paragraph for the paper which tells about a significant feature of your compound and some context as to its importance to the world.

2). Describe how you would characterize the compound. In particular, focus on the coordination of the carbonate ion and the number of phenanthroline ligands. Editor's note: ACS journals require two forms of characterization.

This is a hypothetical problem so you can make up your results and weave a story around those results. However, your results must be within the bounds of the facts given.